

What is Software Engineering

Software engineering is problem solving. But so is medicine, law, architecture, and nearly every other profession. What makes software engineering distinct?

Problem Solving Through Software

Software engineers solve problems using **software as the medium**. We translate real-world problems into software solutions.

This has been true since the beginning:

- When programming meant wiring circuits
- When it meant punching cards
- When it meant copying code from StackOverflow
- And now, when AI can generate code for us

The tools change. The essence doesn't. We identify problems and build software to solve them.

What Makes Software Unique

Software has characteristics that set it apart from other engineering disciplines:

- **Abstract** — You can't touch it. It exists as logic and instructions.
- **Malleable** — It can be changed easily, which is both a strength and a danger.
- **Complex at scale** — Small programs are simple. Real systems have millions of interconnected pieces.
- **The solution becomes the problem** — The code we write to solve problems can itself become a problem (technical debt).

Not About Making Code

The focus has never been "making code." It has always been "providing solutions."

We write code because it's how we deliver solutions. But code is a means, not an end. If we could solve the problem without writing code, that would often be better.

This is why understanding the problem matters more than jumping to implementation. The best engineers spend significant time understanding what they're solving before writing a single line.